I Source and filtering

OmniPath

causal PPI · CollecTRI · KinaseExtra

- directed, signed one way
- both partners single proteins
- ≥2 supporting publications
- sources disagree on sign

11 contradictory pairs removed

II Each protein's typical direction

count what each protein does

to all its OTHER partners

AKT1		usually activates
PTPN1		usually inhibits

only proteins with a clear default are kept (70%+ one direction)

activates · inhibits

III One matched pair per protein

AKT1
usually activates

AKT1 → partner A
answer: stimulates
typical

AKT1 → partner B
answer: inhibits
EXCEPTION

the two partners are matched on number of supporting publications
(median 3 vs 3, Mann–Whitney $P = 0.91$)

IV The resulting benchmark

216 questions · 108 proteins · answers balanced 50/50

protein usually ↓ \ true answer →	inhibits	stimulates
stimulates	70 exception	70 typical
inhibits	38 typical	38 exception

guessing scores 50% · always following the habit scores 50%

Controls

- **120 fabricated pairs**
  real proteins, no documented interaction, matched on connectivity
- **options reversed**
  the same items with "inhibits" listed before "stimulates"
- **40-item label audit**
  checked blind against the cited primary literature

Asked separately

"Across its known direct targets, what is the net regulatory effect of AKT1 at the protein level?"

one probe per protein (n = 108)

Elicits the model's OWN belief about the protein, in a separate session, so it cannot prime the main questions.

Scoring

correct sign	wrong sign	"context dependent"	"not established"
right	wrong	wrong — the model committed	abstention, tracked separately

Accuracy is reported two ways: counting abstentions as incorrect (what a pipeline receives), and on committed answers only (what the model knows).